

# Internal Audit Certificate

Paper III — Linear-Error Clique Partitions of Split Graphs (Erdos #81)

Audit block: Block 04 — Corridor integral packing (Lemma 5.1 & Cor 5.3)

**VERDICT: PASS (E-5.1 180/180, Cor5.3 180/180, basic 372/372)**

## Scope

Exact 0/1 ILP (PuLP/CBC) computation of  $\nu_3(G)$  on 372 systematically generated split graphs, verifying E-5.1 ( $\nu_3 \geq (1/q) \sum C(d_i, 2)$ ) and Corollary 5.3 ( $\Phi \leq n^2/6 + p/2 + (s^2 - 6s + 3)/12$ ) on the applicable instances, plus basic invariants.

## Summary of evidence

- 372/372 instances satisfy  $0 \leq \Phi$  and  $3\nu_3 \leq |E|$ .
- E-5.1: 180/180 applicable instances PASS.
- Corollary 5.3: 180/180 applicable instances PASS.

## Methodology

All numeric checks use exact rational arithmetic (Python fractions) or exact linear/integer programming; symbolic identities are proved with SymPy (simplify(LHS-RHS)=0 or sum-of-squares certificate). The audit re-derives each claim independently of the manuscript's own scripts and of the Lean formalization.

## Reproducibility

Results file: results/corridor\_ILP\_results.txt

SHA-256 (results file): eeb55cbb2b33abcbeae841b6ed2562b1a3ba039c3dfdbe5f0074d2561b9fa7ce

Generated: 2026-07-21 16:38 UTC | Tooling: Python 3.14, SymPy 1.14, SciPy 1.17, PuLP/CBC

Auditor: internal automated audit harness (Claude Code)

*This certificate attests to computational/symbolic verification of the stated finite/closed-form claims at the audited parameter ranges. It complements, and is independent of, the machine-checked Lean 4 / Mathlib formalization.*